



1. Description

1.1. Project

Project Name	foxilloscope
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.18.1
Date	10/05/2026

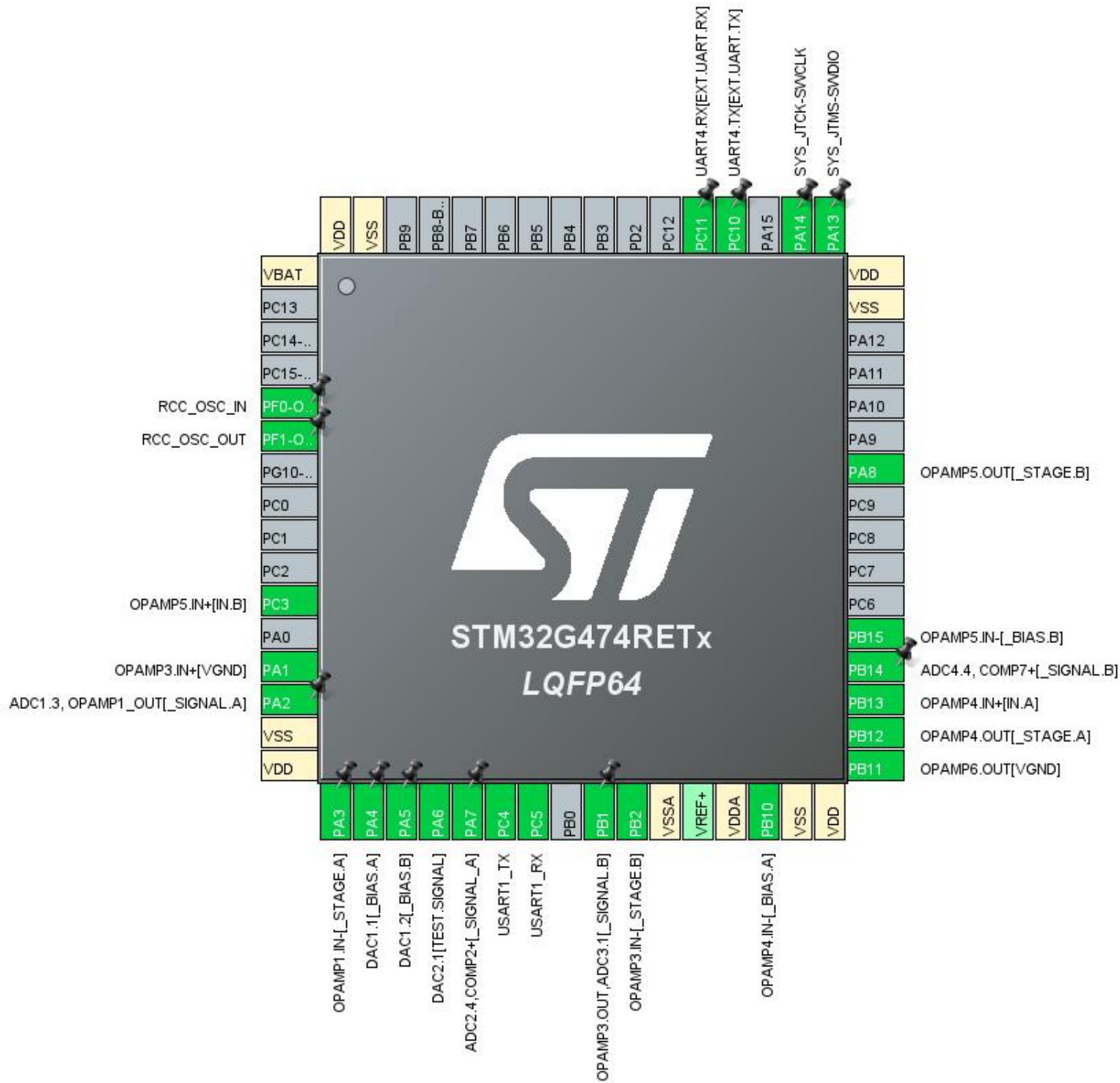
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



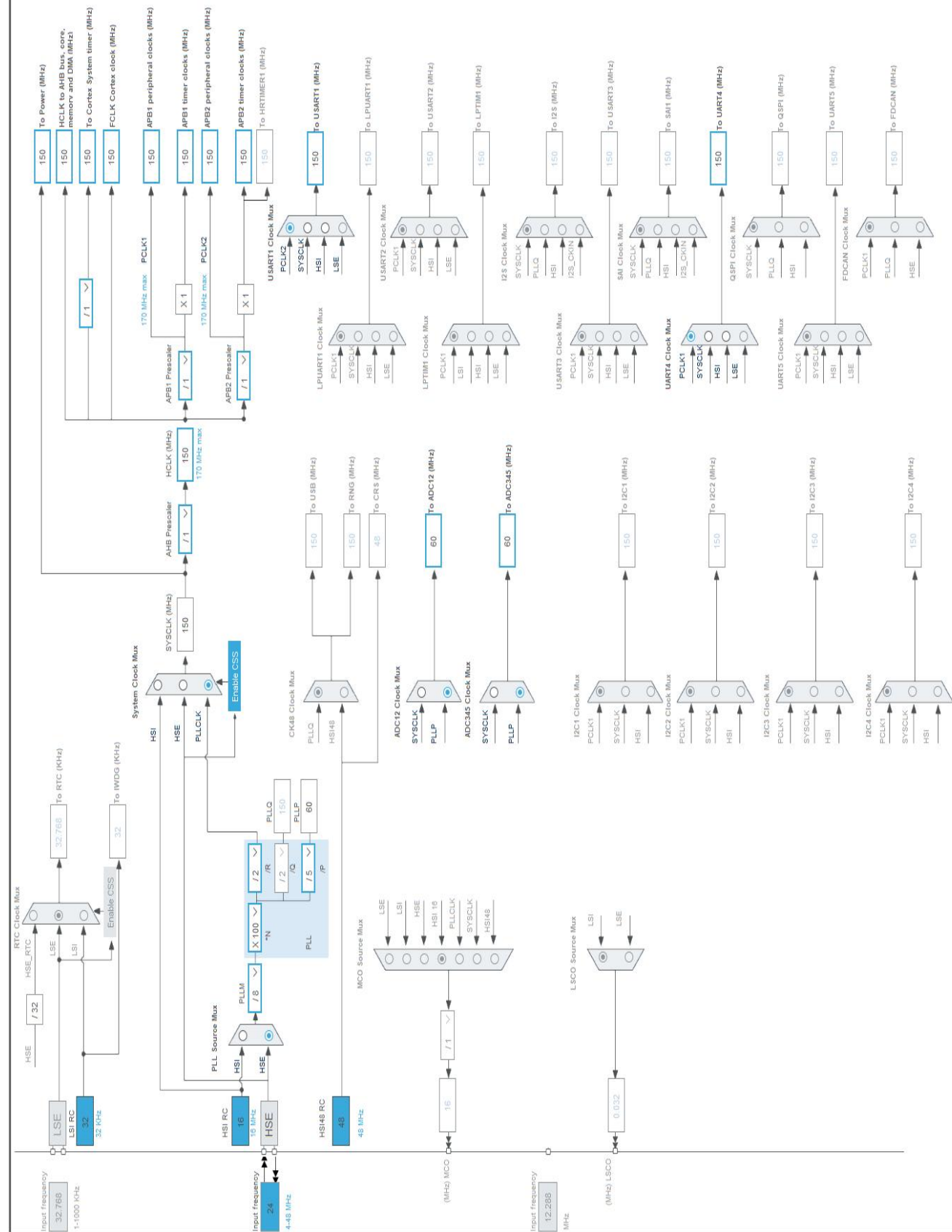
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
5	PF0-OSC_IN	I/O	RCC_OSC_IN	
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	
11	PC3	I/O	OPAMP5_VINP	OPAMP5.IN+[IN.B]
13	PA1	I/O	OPAMP3_VINP	OPAMP3.IN+[VGND]
14	PA2	I/O	ADC1_IN3, OPAMP1_VOUT	ADC1.3, OPAMP1_OUT[SIGNAL.A]
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	OPAMP1_VINM0	OPAMP1.IN-[STAGE.A]
18	PA4	I/O	DAC1_OUT1	DAC1.1[BIAS.A]
19	PA5	I/O	DAC1_OUT2	DAC1.2[BIAS.B]
20	PA6	I/O	DAC2_OUT1	DAC2.1[TEST.SIGNAL]
21	PA7	I/O	COMP2_INP, ADC2_IN4	ADC2.4,COMP2+[SIGNAL_A]
22	PC4	I/O	USART1_TX	
23	PC5	I/O	USART1_RX	
25	PB1	I/O	ADC3_IN1, OPAMP3_VOUT	OPAMP3.OUT,ADC3.1[SIGNAL.B]
26	PB2	I/O	OPAMP3_VINM0	OPAMP3.IN-[STAGE.B]
27	VSSA	Power		
29	VDDA	Power		
30	PB10	I/O	OPAMP4_VINM0	OPAMP4.IN-[BIAS.A]
31	VSS	Power		
32	VDD	Power		
33	PB11	I/O	OPAMP6_VOUT	OPAMP6.OUT[VGND]
34	PB12	I/O	OPAMP4_VOUT	OPAMP4.OUT[STAGE.A]
35	PB13	I/O	OPAMP4_VINP	OPAMP4.IN+[IN.A]
36	PB14	I/O	ADC4_IN4, COMP7_INP	ADC4.4, COMP7+[SIGNAL.B]
37	PB15	I/O	OPAMP5_VINM0	OPAMP5.IN-[BIAS.B]
42	PA8	I/O	OPAMP5_VOUT	OPAMP5.OUT[STAGE.B]
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	
50	PA14	I/O	SYS_JTCK-SWCLK	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
52	PC10	I/O	UART4_TX	UART4.TX[EXT.UART.TX]
53	PC11	I/O	UART4_RX	UART4.RX[EXT.UART.RX]
63	VSS	Power		
64	VDD	Power		

4. Clock Tree Configuration

RIF configuration not available for STM32G474RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

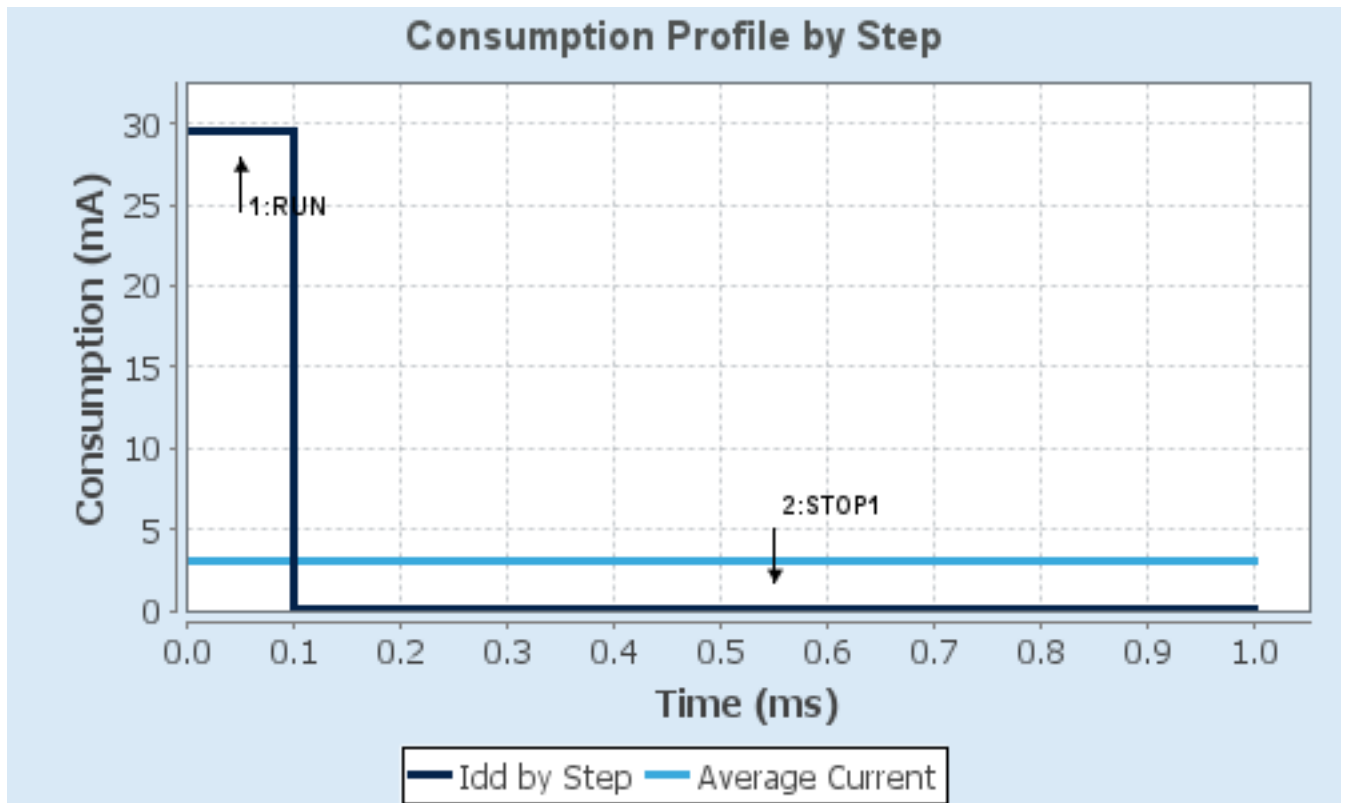
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	foxilloscope
Project Folder	D:\projects\foxilloscope
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	Yes

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_DAC1_Init	DAC1
5	MX_TIM1_Init	TIM1
6	MX_TIM15_Init	TIM15
7	MX_TIM2_Init	TIM2
8	MX_DAC3_Init	DAC3
9	MX_ADC1_Init	ADC1
10	MX_ADC2_Init	ADC2
11	MX_UART4_Init	UART4

Rank	Function Name	Peripheral Instance Name
12	MX_ADC3_Init	ADC3
13	MX_ADC4_Init	ADC4
14	MX_DAC2_Init	DAC2
15	MX_USART1_UART_Init	USART1
16	MX_OPAMP6_Init	OPAMP6
17	MX_OPAMP1_Init	OPAMP1
18	MX_OPAMP4_Init	OPAMP4
19	MX_OPAMP3_Init	OPAMP3
20	MX_OPAMP5_Init	OPAMP5
21	MX_COMP2_Init	COMP2
22	MX_COMP7_Init	COMP7
23	MX_DAC4_Init	DAC4

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN3: IN3 Single-ended

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual Interleaved mode + injected simultaneous mode *

DMA Access Mode

DMA access mode enabled

Delay between 2 sampling phases

7 Cycles *

ADC_Settings:

Clock Prescaler

Asynchronous clock mode divided by 1 *

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Enabled *

Overrun behaviour

Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Timer 2 Trigger Out event *

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 3

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Enable *

Number Of Conversions

1 *

External Trigger Source

Injected Conversion launched by software

External Trigger Conversion Edge

None

Enable Injected Oversampling

Disable

Injected Conversion Mode

None

Injected Queue	Injected Queue Disable
<u>Rank</u>	1
Channel	Channel Vrefint *
Sampling Time	640.5 Cycles *
Offset Number	No offset

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.2. ADC2

IN4: IN4 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Dual Interleaved mode + injected simultaneous mode *
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	7 Cycles *

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1 *
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
<u>Rank</u>	1
Channel	Channel 4

Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.3. ADC3

mode: IN1

3.3.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Dual interleaved mode only *
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	7 Cycles *

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1 *
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Enabled *
Overrun behaviour	Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Timer 2 Trigger Out event *
External Trigger Conversion Edge	Trigger detection on the rising edge
<u>Rank</u>	1

Channel	Channel 1
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false
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Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false
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3.4. ADC4

IN4: IN4 Single-ended

3.4.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Dual interleaved mode only *
DMA Access Mode	DMA access mode enabled
Delay between 2 sampling phases	7 Cycles *

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1 *
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Gain Compensation	0
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
<u>Rank</u>	1
Channel	Channel 4
Sampling Time	2.5 Cycles

Enable Injected Conversions Disable

Enable Analog WatchDog1 Mode	false
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Enable Analog WatchDog2 Mode	false
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Enable Analog WatchDog3 Mode	false
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3.5.1. Parameter Settings:

Level 50mV *

COMP output on GPIO isn't inverted

3.6.1. Parameter Settings:

Level 50mV *

COMP output on GPIO isn't inverted

3.7. DAC1

OUT1 mode: Connected to external pin only

OUT2 mode: Connected to external pin only

3.7.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.8. DAC2

OUT1 mode: Connected to external pin only

3.8.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DAC High Frequency	Mode Automatic
DMA Double Data	Enable *
Signed Format	Disable
Trigger	Timer 15 Trigger Out event *
Trigger2	None
Wave generation mode	Disabled
User Trimming	Factory trimming

3.9. DAC3

mode: OUT1 mode

mode: OUT2 mode

3.9.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Disable *
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Disable *
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.10. DAC4

mode: OUT1 mode

3.10.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None

User Trimming

Factory trimming

3.11. OPAMP1

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS-DAC3_OUT1-INP

3.11.1. Parameter Settings:

Basic Parameters:

Power Mode

High Speed *

PGA Gain

2 or -1

User Trimming

Disable

3.12. OPAMP3

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS

3.12.1. Parameter Settings:

Basic Parameters:

Power Mode

High Speed *

PGA Gain

2 or -1

User Trimming

Disable

3.13. OPAMP4

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS

3.13.1. Parameter Settings:

Basic Parameters:

Power Mode

High Speed *

PGA Gain

2 or -1

User Trimming

Disable

3.14. OPAMP5

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS

3.14.1. Parameter Settings:

Basic Parameters:

Power Mode	Normal
PGA Gain	2 or -1
User Trimming	Disable

3.15. OPAMP6

Mode: Follower-DAC3_OUT1-INP

3.15.1. Parameter Settings:

Basic Parameters:

Power Mode	High Speed *
User Trimming	Disable

3.16. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.16.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
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3.17. SYS

Debug: Serial Wire

VREFBUF Mode: External voltage reference

Timebase Source: TIM7

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.18. TIM1

Slave Mode: External Clock Mode 1

Trigger Source: ITR1

Channel1: PWM Generation No Output

mode: One Pulse Mode

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Output Compare (OC1REF) *
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
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BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	99 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Set *

3.19. TIM2

Slave Mode: Gated Mode

Trigger Source: ITR0

Clock Source : Internal Clock

3.19.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	39 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	19 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Slave Mode Controller

Gated Mode

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)

Disable (Trigger input effect not delayed)

Trigger Event Selection TRGO

Update Event *

3.20. TIM15

mode: Clock Source

3.20.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)

30 *

Counter Mode

Up

Dithering

Disable

Counter Period (AutoReload Register - 16 bits value)

16 *

Internal Clock Division (CKD)

No Division

Repetition Counter (RCR - 8 bits value)

0

auto-reload preload

Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)

Disable (Trigger input effect not delayed)

Trigger Event Selection

Update Event *

3.21. UART4

Mode: Asynchronous

3.21.1. Parameter Settings:

Basic Parameters:

Baud Rate

460800 *

Word Length

8 Bits (including Parity)

Parity

None

Stop Bits

1

Advanced Parameters:

Data Direction

Receive and Transmit

Over Sampling

16 Samples

Single Sample

Disable

ClockPrescaler

1

Fifo Mode

FIFO mode enable *

Txfifo Threshold	half full configuration *
Rxfifo Threshold	1 eighth full configuration
Advanced Features:	
Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.22. USART1

Mode: Asynchronous

3.22.1. Parameter Settings:

Basic Parameters:

Baud Rate	460800 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Enable *
Txfifo Threshold	half full configuration *
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.23. FREERTOS

Interface: CMSIS_V2

3.23.1. Config parameters:

API:

FreeRTOS API CMSIS v2

Versions:

FreeRTOS version 10.3.1

CMSIS-RTOS version 2.1.3

MPU/FPU:

ENABLE_MPU Disabled

ENABLE_FPU Disabled

Kernel settings:

USE_PREEMPTION Enabled

CPU_CLOCK_HZ SystemCoreClock

TICK_RATE_HZ 1000

MAX_PRIORITIES 56

USE_SB_COMPLETED_CALLBACK 0

USE_MINI_LIST_ITEM 1

MINIMAL_STACK_SIZE 128

MAX_TASK_NAME_LEN 16

USE_16_BIT_TICKS Disabled

IDLE_SHOULD_YIELD Enabled

USE_MUTEXES Enabled

USE_RECURSIVE_MUTEXES Enabled

USE_COUNTING_SEMAPHORES Enabled

QUEUE_REGISTRY_SIZE 8

USE_APPLICATION_TASK_TAG Disabled

ENABLE_BACKWARD_COMPATIBILITY Enabled

USE_PORT_OPTIMISED_TASK_SELECTION Disabled

USE_TICKLESS_IDLE Disabled

USE_TASK_NOTIFICATIONS Enabled

RECORD_STACK_HIGH_ADDRESS Disabled

Memory management settings:

Memory Allocation Dynamic / Static

TOTAL_HEAP_SIZE **30720 ***

Memory Management scheme heap_4

Hook function related definitions:

USE_IDLE_HOOK	Enabled *
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.23.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled

xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.23.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA2	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	ADC1.3, OPAMP1_OUT[SIGNAL. A]
ADC2	PA7	ADC2_IN4	Analog mode	No pull-up and no pull-down	n/a	ADC2.4,COMP2+[SIGNAL L_A]
ADC3	PB1	ADC3_IN1	Analog mode	No pull-up and no pull-down	n/a	OPAMP3.OUT,ADC3.1[SI GNAL.B]
ADC4	PB14	ADC4_IN4	Analog mode	No pull-up and no pull-down	n/a	ADC4.4, COMP7+[SIGNAL.B]
COMP2	PA7	COMP2_INP	Analog mode	No pull-up and no pull-down	n/a	ADC2.4,COMP2+[SIGNAL L_A]
COMP7	PB14	COMP7_INP	Analog mode	No pull-up and no pull-down	n/a	ADC4.4, COMP7+[SIGNAL.B]
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	DAC1.1[BIAS.A]
	PA5	DAC1_OUT2	Analog mode	No pull-up and no pull-down	n/a	DAC1.2[BIAS.B]
DAC2	PA6	DAC2_OUT1	Analog mode	No pull-up and no pull-down	n/a	DAC2.1[TEST.SIGNAL]
OPAMP1	PA2	OPAMP1_VOUT	Analog mode	No pull-up and no pull-down	n/a	ADC1.3, OPAMP1_OUT[SIGNAL. A]
	PA3	OPAMP1_VINM0	Analog mode	No pull-up and no pull-down	n/a	OPAMP1.IN-[STAGE.A]
OPAMP3	PA1	OPAMP3_VINP	Analog mode	No pull-up and no pull-down	n/a	OPAMP3.IN+[VGND]
	PB1	OPAMP3_VOUT	Analog mode	No pull-up and no pull-down	n/a	OPAMP3.OUT,ADC3.1[SI GNAL.B]
	PB2	OPAMP3_VINM0	Analog mode	No pull-up and no pull-down	n/a	OPAMP3.IN-[STAGE.B]
OPAMP4	PB10	OPAMP4_VINM0	Analog mode	No pull-up and no pull-down	n/a	OPAMP4.IN-[BIAS.A]
	PB12	OPAMP4_VOUT	Analog mode	No pull-up and no pull-down	n/a	OPAMP4.OUT[STAGE.A]
	PB13	OPAMP4_VINP	Analog mode	No pull-up and no pull-down	n/a	OPAMP4.IN+[IN.A]
OPAMP5	PC3	OPAMP5_VINP	Analog mode	No pull-up and no pull-down	n/a	OPAMP5.IN+[IN.B]
	PB15	OPAMP5_VINM0	Analog mode	No pull-up and no pull-down	n/a	OPAMP5.IN-[BIAS.B]
	PA8	OPAMP5_VOUT	Analog mode	No pull-up and no pull-down	n/a	OPAMP5.OUT[STAGE.B]
OPAMP6	PB11	OPAMP6_VOUT	Analog mode	No pull-up and no pull-down	n/a	OPAMP6.OUT[VGND]
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
UART4	PC10	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	UART4.TX[EXT.UART.TX]

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC11	UART4_RX	Alternate Function Push Pull	Pull-down *	Low	UART4.RX[EXT.UART.RX]
USART1	PC4	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC5	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	

4.2. DMA configuration

DMA request	Stream	Direction	Priority
MENTOMEM	DMA1_Channel2	Memory To Memory	Low
ADC1	DMA1_Channel5	Peripheral To Memory	Low
MENTOMEM	DMA1_Channel6	Memory To Memory	Low
ADC3	DMA1_Channel3	Peripheral To Memory	Low
UART4_TX	DMA1_Channel7	Memory To Peripheral	Low
USART1_TX	DMA1_Channel4	Memory To Peripheral	Low
DAC2_CH1	DMA1_Channel1	Memory To Peripheral	Low

MENTOMEM: DMA1_Channel2 DMA request Settings:

Mode: Normal
 Src Memory Increment: **Enable ***
 Dst Memory Increment: **Enable ***
 Src Memory Data Width: **Word ***
 Dst Memory Data Width: **Word ***

ADC1: DMA1_Channel5 DMA request Settings:

Mode: **Circular ***
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: **Word ***
 Memory Data Width: **Word ***

MENTOMEM: DMA1_Channel6 DMA request Settings:

Mode: Normal
 Src Memory Increment: **Enable ***
 Dst Memory Increment: **Enable ***
 Src Memory Data Width: **Word ***
 Dst Memory Data Width: **Word ***

ADC3: DMA1_Channel3 DMA request Settings:

Mode: **Circular ***
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: **Word ***
Memory Data Width: **Word ***

UART4_TX: DMA1_Channel7 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART1_TX: DMA1_Channel4 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

DAC2_CH1: DMA1_Channel1 DMA request Settings:

Mode: **Circular ***
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Word
Memory Data Width: **Word ***

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	5	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel6 global interrupt	true	5	0
DMA1 channel7 global interrupt	true	5	0
TIM1 capture compare interrupt	true	5	0
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	true	5	0
UART4 global interrupt / UART4 wake-up interrupt through EXTI line 34	true	5	0
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	true	15	0
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	true	0	0
COMP7 interrupt through EXTI line 33	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
DMA1 channel1 global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM2 global interrupt	unused		
ADC3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
ADC4 global interrupt		unused	
FPU global interrupt		unused	

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
DMA1 channel7 global interrupt	false	true	true
TIM1 capture compare interrupt	false	true	true
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	false	true	true
UART4 global interrupt / UART4 wake-up interrupt through EXTI line 34	false	true	true
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	false	true	true
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	false	true	true
COMP7 interrupt through EXTI line 33	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware								
FREERTOS								
System Core	Analog		Timers	Connectivity	Multimedia	Security	Computing	Utilities
DMA	ADC1	ADC2	TIM1	UART4				
GPIO	ADC3	ADC4	TIM2	USART1				
NVIC	COMP2	COMP7	TIM15					
RCC	DAC1	DAC2						
SYS	DAC3	DAC4						
	OPAMP1	OPAMP3						
	OPAMP4	OPAMP5						
	OPAMP6							

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32c5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-

use-tips-and-recommendations-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smmps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5978-introduction-to-mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2548-introduction-to-dma-controller-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf

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the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an3126-how-to-use-dac-for-audio-and-waveform-generation-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-how-to-use-usart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4286-how-to-use-spi-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4760-introduction-to-quadspi-interface-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00431551.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf